

Projective Spaces: Let V be a complex finite dim vector space. and let $\mathbb{P}(V) := \{1\text{-D subspaces of } V\}$
 We call $\mathbb{P}(V)$ Projective space of a vector space V .

If $V := \mathbb{C}^{n+1}$, then $\mathbb{P}_n(\mathbb{C}) := \mathbb{P}(\mathbb{C}^{n+1})$ which is an n -dim complex projective space.

$\mathbb{CP}^n$ admits a topology by noting that $\exists$ a projection $\text{pr}: \mathbb{C}^{n+1} \setminus \{0\} \rightarrow \mathbb{CP}^n$

$\text{pr}(X) = \lambda X$ for $X \neq 0$, where $\lambda \in \mathbb{C}^*$

here $\text{pr}(X) \equiv [X] = [\lambda X]$

• $\mathbb{CP}^n$ has the Quotient topology of $\mathbb{C}^{n+1} / \mathbb{C}^*$

$U \subset \mathbb{CP}^n$ is open iff $\text{pr}^{-1}(U)$ is open in $\mathbb{C}^{n+1}$.

$\phi_i: (w_1, \dots, w_n) \in \mathbb{C}^n \mapsto [w_1, \dots, w_i, 1, w_{i+1}, \dots, w_n]$

$\mathbb{CP}^1 \cong S^2$

$\mathbb{CP}^1$ is the one-point compactification of $\mathbb{C} \rightarrow \mathbb{C} \cup \{\infty\}$

Grassmannian Manifolds: Let V be a fixed n -dim complex vector space. The Grassmannian mfd of k -dim subspaces of V

$G_k(V) := \{k\text{-dim subspaces of } V\}$

$G_k(V)$ has the structure of a complex mfd of dim

$\dim(G_k(V)) = k(n-k)$

• Note: $k=1$ $G_1(V) = \mathbb{P}(V) \cong \mathbb{CP}^{n-1}$

To understand the structure of $G_k(V)$ we must generalize the notion of homogeneous coords

$X^T = \lambda Y^T$. In particular, let $M_{k,n}(\mathbb{C})$ be the

Set of $k \times n$ matrices of rank k ($k \leq n$)

$A, B \in M_{k,n}(\mathbb{C})$ and define $A \sim B$

$B = gA$, $g \in GL(k, \mathbb{C})$. Thus the above space is the Quotient
 $G_{k,n}(\mathbb{C}) \cong M_{k,n}(\mathbb{C}) / GL(k, \mathbb{C})$

Projective Algebraic Mod:

The common zeros of a set of homogeneous polynomials are a compact submfld of $\mathbb{CP}^n$ called an algebraic variety N . Let $\{P_\alpha\}_{\alpha \in I}$ be the set of homogeneous polynomials in $\mathbb{C}^{n+1}$ s.t.

$$N := \{[x^0, \dots, x^n] \mid P_\alpha(x^0, \dots, x^n) = 0, \forall \alpha \in \{1, \dots, r\}\}$$

* Note, let P_i be a polynomial of degree d , then
 $P_i(\lambda x^0, \dots, \lambda x^n) = \lambda^d P_i(x^0, \dots, x^n)$

So P_i are defined in $\mathbb{CP}^n$. To see this simply note

$$P_i(x^0, \dots, x^n) = 0 \Rightarrow P([x^0, \dots, x^n]) = 0$$

ex: Let $P_i(x^0, x^1, x^2) := (x^0)^2 + (x^1)^2 + (x^2)^2$
 $\Rightarrow N = \{[x^0, x^1, x^2] \in \mathbb{CP}^2 \mid P_i(x^0, x^1, x^2) = 0\}$

In $N \cap U_{(0)}$ we can define inhomogeneous coords:

$$1 + [z_{(0)}^1]^2 + [z_{(0)}^2]^2 = 0.$$

Quartic in $\mathbb{CP}^n$: Let q_{ij} be a complex symm. matrix and define the Bi-linear

$$q(Z, W) = Z^T q_{ij} W^j, \quad Z, W \in \mathbb{C}^{n+1}$$

Def: q is nondegenerate if $q(Z, W) = 0 \quad \forall W \in \mathbb{C}^{n+1}$ implies $Z = 0$. At the level of matrices this

means $\det(q_{ij}) \neq 0$

- The only invariant of q , up to complex linear change of variables is the rank: the # of nonzero diagonal entries in the diagonalization.

$$Q \equiv \{[X] \in \mathbb{CP}^n \mid q(X, X) = 0\}$$

This is a projective algebraic variety in $\mathbb{CP}^n$.

Note, to find such varieties:

$$\frac{\partial q}{\partial X^i} = 2 q_{i\bar{j}} X^{\bar{j}} = 0$$

So the singular set of Q is given by

$$\ker(q) : \mathbb{C}^{n+1} \rightarrow \mathbb{C}^{n+1}$$

- Q is nonsingular (at all pts) iff q is nondegenerate: i.e. has maximal rank $= n+1$

Twistor Manifolds in Correspondences:

We now generalize the previous notions:

- Let V be a fixed complex vect. space of dim n and let $\{d_1, \dots, d_m\}$ be a seq. of positive integers satisfying $1 \leq d_1 < d_2 < \dots < d_m \leq n$. Then

$$F_{d_1, \dots, d_m}(V) := \{(S_1, \dots, S_m) \mid S_i \text{ are subspaces of } V \text{ of dim } d_i, S_1 \subset S_2 \subset \dots \subset S_m\}$$

which are known as a flag-manifold of type $(d_1, \dots, d_m)$.

*We use the above flag manifold to define the twistor geometry PII. Similarly, fixing a twistor space PII is similar to fixing a Hilbert Space of a Quantum theory.

- We associate A family of Flag manifolds w/ PII. Let

$$V = \mathbb{P}T\mathbb{P}$$

$\dim(V) = 4$ so we can only build the flags

$F_1, F_2, F_3, F_{12}, F_{13}, F_{23}, F_{123}$ etc...

* Now, consider the following flags

$$[F_{12}, F_1, F_2]$$

$$\begin{array}{ccc} & F_{12} & \\ \mu \swarrow & & \searrow \nu \\ F_1 & & F_2 \end{array} \quad \begin{array}{l} \mu(S_1, S_2) = S_1 \\ \nu(S_1, S_2) = S_2 \end{array}$$

μ, ν : Both surjective holomorphic mappings of maximal rank
i.e. they are projections of the bundle.

• This is a Double fibration

We have a natural correspondence

$$F_1 \xrightarrow{c} F_2$$

$$c(p) := \nu \circ \mu^{-1}(p)$$

Here $F_{12} \cong \mathbb{P}\mathbb{S}$ projective spinor space

$$\left. \begin{array}{l} \mathbb{P}T\mathbb{P} := F_1 = F_1(\pi) \cong \mathbb{P}\mathbb{P}^3 \\ \mathbb{M}\mathbb{Q} := F_2 = F_2(\pi) \cong G_{2,4}(\mathbb{Q}) \\ \mathbb{P}\mathbb{S} := F_{12} = F_{12}(\pi) \end{array} \right\}$$

